Skyline Delivery Fleet

Week 7 Assignment - Problem 5 (Intermediate)

DeliveryDrone and ScoutDrone are siblings under Drone. Only DeliveryDrone implements Trackable; GroundRobot implements Trackable independently. instanceof safely checks the capability.

Java solution

```
abstract class Drone {
    public abstract String fly();
}
interface Trackable {
    String getLocation();
}
class DeliveryDrone extends Drone implements Trackable {
    private final String id;
    DeliveryDrone(String id) { this.id = id; }
    public String fly() { return "Delivery drone " + id + " flying"; }
    public String getLocation() { return id + " at Sector 4"; }
}
class ScoutDrone extends Drone {
    private final String id;
    ScoutDrone(String id) { this.id = id; }
    public String fly() { return "Scout drone " + id + " flying"; }
}
class GroundRobot implements Trackable {
    private final String id;
    GroundRobot(String id) { this.id = id; }
    public String getLocation() { return id + " at Sector 4"; }
}
public class Main {
    static String getLocationIfTrackable(Object o) {
        if (o instanceof Trackable) {
            return ((Trackable) o).getLocation();
        }
        return "Tracking not available";
    }
    public static void main(String[] args) {
        System.out.println(getLocationIfTrackable(
            new DeliveryDrone("DR-1")));
        System.out.println(getLocationIfTrackable(
            new ScoutDrone("SC-1")));
        System.out.println(getLocationIfTrackable(
            new GroundRobot("GR-1")));
    }
}
```

Sample output

```
DR-1 at Sector 4
Tracking not available
GR-1 at Sector 4
```

Note: Trackability depends on implementing Trackable, not on being a Drone.